

Replay-Aware CVRP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: cvrp_no_replay.
- Best CVRPLIB holdout gap: cvrp_no_replay.
- Best synthetic holdout gap: cvrp_no_replay.

Run Metadata

- run_name: run_20260513_190437_a
- started_at_local: 2026-05-13 19:04:37
- finished_at_local: 2026-05-13 19:07:27
- duration_hhmm: 00:03
- duration_seconds: 170.049
- seed_offset: 0
- replicate_label: a
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
cvrp_no_replay	none	score_only	False	0.243302	0.005376	0.137557	0.906959	0.58	0.001157
cvrp_random_replay	random	score_only	False	0.274255	0.02481	0.163391	0.779781	0.78	-0.031245
cvrp_failure_replay	failure	score_only	False	0.25085	0.005376	0.14175	0.0	0.58	0.0
cvrp_failure_replay_compression	failure	novelty_gate	True	0.261192	0.005376	0.147496	0.0	0.58	0.0

Condition Notes

cvrp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.243302, synthetic 0.005376, combined 0.137557.

- Accepted-epoch count 3, mean accepted code novelty 0.906959, and final complexity 0.58.
- Adaptation efficiency 0.001157 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.243302 across 5 instances; family means: A=0.112565, B=0.255548, E=0.285988, P=0.306859.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 593, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.29476}, {"best_known_cost": 937, "cost": 1064, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.135539}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.274255, synthetic 0.02481, combined 0.163391.
- Accepted-epoch count 2, mean accepted code novelty 0.779781, and final complexity 0.78.
- Adaptation efficiency -0.031245 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.274255 across 5 instances; family means: A=0.165794, B=0.250139, E=0.414587, P=0.290614.
- Panel synthetic_holdout mean gap 0.02481 across 4 instances; family means: alternating_belt=0.049043, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.050197.
- Worst recent training cases: [{"best_known_cost": 784, "cost": 1017, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.297194}, {"best_known_cost": 458, "cost": 567, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.237991}, {"best_known_cost": 937, "cost": 1142, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.218783}]

cvrp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.25085, synthetic 0.005376, combined 0.14175.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.58.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.25085 across 5 instances; family means: A=0.143979, B=0.254873, E=0.293666, P=0.306859.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 604, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.318777}, {"best_known_cost": 937, "cost": 1064, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.135539}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: CVRPLIB 0.261192, synthetic 0.005376, combined 0.147496.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.58.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.261192 across 5 instances; family means: A=0.143979, B=0.254873, E=0.34357, P=0.308664.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 585, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.277293}, {"best_known_cost": 937, "cost": 1073, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.145144}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

Judge Appendix

Summary of Replay-Aware CVRP Benchmark Results

Condition	Final C VRPLI B Gap	Final Sy nthetic Gap	Final Tr ansfer Gap	Mean Code Novelt y	Replay Mode	Co mp lex ity	Adaptati on Efficie ncy	Notes
cvrp_no_repla y	**0.243 3**	**0.0054 **	**0.137 6**	0.907	None	0.5 8	0.00116	Best performance overall on transfer and held-out benchmarks.
cvrp_random_r eplay	0.2743	0.0248	0.1634	0.780	Random replay	0.7 8	-0.0312	Higher gaps despite moderate code novelty; worse transfer.
cvrp_failure_rep lay	0.2509	0.0054	0.1418	0.0	Failure replay	0.5 8	0.0	Performance slightly worse than no replay; zero code novelty.
cvrp_failure_rep lay_compressio n	0.2612	0.0054	0.1475	0.0	Failure + compress ion	0.5 8	0.0	Compression pressure present; no improvements over failure replay.

Conservative Interpretation

- ****Transfer Metrics (Held-out CVRPLIB & Synthetic Holdout Gaps):****

The ****no replay**** condition consistently yields the lowest mean optimality gaps on both held-out CVRPLIB (0.2433) and synthetic benchmarks (0.0054), as well as the lowest transfer gap (0.1376). This supports superior generalization and transfer capability without replay.

- ****Replay Modes and Impact:****
- ****Random replay**** results in the worst transfer and held-out gaps, accompanied by the lowest adaptation efficiency, despite a relatively high code novelty (~0.78).
- ****Failure replay**** (with and without compression pressure) yields slightly better transfer gaps than random replay but still worse than no replay. Notably, failure replay conditions show zero code novelty, indicating code reuse rather than innovation.
- ****Code Novelty vs. Transfer:****

High code novelty (no replay) correlates here with superior transfer performance. Conversely, failure replay conditions show zero code novelty but do not improve transfer gaps compared to no replay. Random replay sees a drop in code novelty relative to no replay and results in decreased transfer performance.

- **Compression Pressure:**

The addition of compression-aware replay (failure + compression) does not improve transfer gaps or held-out performance and maintains zero code novelty, suggesting no benefit from compression in this context.

Key Takeaways

- The **no replay** condition is optimal for transfer, yielding the best held-out CVRPLIB gaps (~24.3%), synthetic gaps (~0.5%), and transfer gaps (~13.8%).
- Introducing replay (random or failure-based) reduces transfer performance despite sometimes lower or equal code novelty, indicating replay harms or fails to enhance transfer robustness in this setting.
- Code novelty decreases under replay conditions; however, lack of code novelty does not translate into better transfer. Thus, no evidence supports equating lexical novelty alone with improved algorithmic invention or transfer.
- Compression-aware replay does not provide transfer benefits here and retains zero code novelty.
- Overall, transfer improvements correlate positively with no replay, moderate complexity (~0.58), and higher code novelty.

Recommendations for Benchmark Suite Usage

- Prioritize the **no replay** condition results when evaluating transfer and generalization performance.
- Treat replay-inclusive conditions (random, failure, compression) as inferior baselines in terms of transfer despite potentially different exploratory behaviors.
- Avoid interpreting code novelty without supporting transfer metrics as evidence of algorithmic novelty or invention.
- Consider analyzing why replay hurts transfer—possibly preventing exploration or overfitting to replayed failures—and investigate this in future work.